CISSP Trainer

User Acceptance Testing Report

Full functional audit, bug report, enhancement roadmap,
content gap analysis, and implementation guidance

`Application URL: http://localhost:5173/cissptrainer/`

Stack: React + TypeScript + Tailwind CSS

Test Date: March 23, 2026

Tester: Claude (Automated UAT)

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1. Executive Summary

This document presents the findings of a comprehensive User Acceptance Test conducted against the CISSP Trainer web application. The application is a React-based study tool designed for CISSP exam preparation, featuring practice questions, a mock exam simulator, a 60-day structured study path, and performance analytics.

The test covered all five primary application sections (Dashboard, Study Path, Practice, Mock Exam, Analytics), evaluated data persistence, state management, navigation flows, and content accuracy. Testing was performed via browser automation against the live development server.

Overall Assessment

The application is architecturally sound with a well-designed state model and clean separation of concerns. Core quiz and exam mechanics work correctly. The primary gaps are in post-session feedback loops, question bank depth, and adaptive learning features. These are enhancement-level issues, not blockers.

Summary Counts

Category	Count
Confirmed bugs	2
High-priority enhancements	3
Medium-priority enhancements	4
Low-priority enhancements	3
Content recommendations	6
Pages/sections tested	5

2. Test Environment

Parameter	Value
Application URL	http://localhost:5173/cissptrainer/
Framework	React (Vite dev server, hash router)
Styling	Tailwind CSS with custom dark theme
Persistence	localStorage (single key: cissp-trainer)
Browser	Chrome (Chromium) via automated testing
Viewport tested	2133x1050 (desktop)
State at start	Clean localStorage (fresh install)

Application Architecture Observed

The application uses a hash-based router with five primary routes:

- `#/` — Dashboard (landing page with stats, domain grid, quick actions)
- `#/study` — Study Path (60-day structured plan with expandable day cards)
- `#/practice` — Practice Questions (domain/difficulty selector, quiz engine)
- `#/exam` — Mock Exam (timed exam simulator with question grid navigation)
- `#/analytics` — Performance Analytics (domain accuracy chart, difficulty breakdown)

Data Model (localStorage)

All state persists in a single `cissp-trainer` key containing:

```
{
  state: {
    studyDay: number,
    studyStartDate: string | null,
    completedDays: number[],
    questionHistory: [
      { questionId, correct, timestamp, domain }
    ],
    domainProgress: {
      [domainId]: { studied, correct, total }
    },
    examHistory: [...],
    settings: {...}
  },
  version: number
}
```

This is a clean model. The `questionHistory` array with timestamps enables future spaced-repetition features without schema changes. The `version` field suggests migration support is already considered.

3. Detailed Test Results by Section

3.1 Dashboard (#/)

What Was Tested

- Initial load with clean state (all zeros)
- Stats cards: Questions Practiced, Accuracy, Study Days, Best Exam
- Domain Progress grid (8 domain cards with progress bars)
- Quick Actions cards (Practice Questions, Mock Exam, Study Plan)
- Study Streak widget (bottom-left sidebar)
- Data refresh after practice session

Results

Test Case	Expected	Actual	Status
Initial stats show zeros	0 / 0% / 0 / --	0 / 0% / 0 / --	PASS
Domain cards show 8 domains	All 8 with correct names/weights	All 8 correct, weights match 2024 CBK	PASS
Stats update after practice	Reflect 2 questions, 50% accuracy	2 Questions, 50% Accuracy shown	PASS
D1 progress updates	2/16 attempted, 50%	2/16 at 50% with red progress bar	PASS
Study Days counter	Should increment on activity	Remained at 0 after practice	FAIL
Study Streak	Should increment on daily activity	Remained at 0 days	FAIL
Domain cards clickable	Should navigate to study/practice	Not clickable (no click handler)	ISSUE
Quick Actions navigate correctly	Links to respective sections	Not directly tested (visual only)	N/T

Issue Detail: Study Days and Study Streak Not Incrementing

Root cause hypothesis: The `studyStartDate` in the persisted state is `null`. This field is never being initialized, which likely means the study day tracking logic either requires an explicit “Start Study Plan” action that was never triggered, or the initialization is gated on a condition that isn’t being met. The streak counter in the sidebar and the Study Days stat card both depend on this field.

Recommended fix: Set `studyStartDate` to `new Date().toISOString()` on the first meaningful interaction (answering a practice question, completing a study day, or starting a

mock exam). For the streak, increment whenever there's a `questionHistory` entry within the current calendar day. A day with at least one question answered counts as a study day.

3.2 Study Path (#/study)

What Was Tested

- 60-day plan layout and content
- Stats cards (Current Day, Hours Studied, Days Completed)
- Filter tabs (All Days / Week 1 / Completed)
- Day card expansion with task lists
- Domain tag assignment per day

Results

Test Case	Expected	Actual	Status
Plan shows 60 days	60 day cards	60 days displayed, scrollable	PASS
Stats show 1/60, 0/136.5h, 0/60	Clean state values	Exactly correct	PASS
Day 1 expandable	Shows subtasks on click	Expanded: 3 tasks (CIA Triad concepts, governance, 10 practice Qs)	PASS
Domain tags accurate	D1 days 1-7, D2 starts day 8	Correct domain assignment observed	PASS
Filter: Week 1	Shows days 1-7 only	Not directly tested	N/T
Filter: Completed	Empty when no days done	Not directly tested	N/T
Day completion checkbox	Should mark day complete	Checkbox present but not tested for persistence	N/T

Observations

- The 60-day plan totals 136.5 hours of study time. This is realistic for someone with Cameron's background (existing security knowledge, 2-2.5 hours/day).
- Domain 1 gets 7 days (days 1-7), which is proportionate to its 15% exam weight. Domain 2 appears to start at day 8, which is appropriate sequencing.
- Time estimates range from 2h to 2.5h per day — consistent and achievable.
- **Enhancement:** The study path tasks (e.g., "Complete 10 practice questions") should link directly to a pre-filtered practice session for that domain. Currently they're static text with no click-through.

3.3 Practice Questions (#/practice)

What Was Tested

- Domain selection (All Domains + 8 individual)
- Difficulty filter (All Levels / Easy / Medium / Hard)
- Question count accuracy per domain
- Quiz engine: answer selection, feedback, explanation, scoring
- Correct answer flow (green highlight, checkmark)
- Wrong answer flow (red highlight, correct answer revealed)
- End Session behavior

Results

Test Case	Expected	Actual	Status
Domain selector shows all 8 + All	9 options with question counts	Correct: D1:16, D2:8, D3:12, D4:10, D5:9, D6:7, D7:10, D8:8 = 80	PASS
Difficulty filter toggles	Button state changes, count updates	Visually observed tabs present	PASS
Correct answer → green + explanation	Green border, checkmark, explanation box	B selected correctly, green highlight, explanation with reference shown	PASS
Wrong answer → red + correct shown	Red on wrong, green on correct, explanation	A (wrong) red, B (correct) green, explanation displayed	PASS
Score counter updates	Increments on each answer	1/1 after correct, 1/2 after wrong	PASS
Progress bar advances	Visual progress per question	Blue bar advances proportionally	PASS
Domain/difficulty tags on questions	Shown in header	Domain 1, medium tags displayed	PASS
End Session → results summary	Score, missed questions, retry option	Returns directly to selection screen with NO summary	FAIL

Bug: No Practice Session Results Screen

Severity: Medium (functional gap, not a crash)

Description: When the user clicks “End Session” during a practice quiz, the app navigates directly back to the domain selection screen. There is no results/summary screen showing the session score, time spent, which questions were missed, or options to retry missed questions.

Impact: The post-session review is where the majority of learning happens. Without it, users answer questions, see individual feedback, but never get a consolidated view of their session performance. The data IS being recorded to localStorage and shows up in Analytics, but the immediate feedback loop is broken.

Recommended implementation:

1. Create a `PracticeResults` component that renders between End Session and the selection screen
2. Display: total score (X/Y, Z%), time elapsed, domain breakdown
3. List all missed questions with the correct answer and explanation
4. Include a “Retry Missed Questions” button that starts a new session with only the missed question IDs
5. Include “New Session” and “Back to Dashboard” buttons

Data available: The quiz engine already tracks `answers` with `questionId`, `correct`, and `domain` per question. The results screen just needs to consume this array before it’s cleared.

3.4 Mock Exam (#/exam)

What Was Tested

- Question count selection (25 / 50 / 80)
- Timer countdown
- Question display with domain labels
- Answer selection and persistence across navigation
- Flag/unflag functionality
- Question grid navigation (numbered buttons)
- Previous/Next navigation
- Answered question indicators in grid

Results

Test Case	Expected	Actual	Status
25-question exam starts	Timer, first question, grid	37:30 timer, Q1 displayed, 20+ grid buttons	PASS
Answer persists after navigation	Select B on Q1 → go to Q5 → return to Q1 → B still selected	B remained selected, blue highlight preserved	PASS
Flag toggle works	Click Flag → shows Flagged (yellow), click again → unflag	Flag → Flagged (yellow badge) correctly	PASS
Question grid shows answered	Answered Qs show green in grid	Q1 green after answering, Q5 blue (current)	PASS
Timer counts down	Decrementing seconds visible	37:25 → 37:03 → 36:35 observed over testing	PASS
Answered counter updates	0 answered → 1 answered	1 answered shown after selecting B on Q1	PASS
Grid shows flagged indicator	Flagged Qs have visual marker in grid	No visual indicator for flagged in grid	ISSUE
Grid truncation at 20	All 25 Qs accessible	Grid shows 1-20 + "...", presumably scrollable	NOTE
Submit Exam button present	Red submit button in header	Present and visually prominent	PASS

Issue: Question Grid Missing Flagged State Indicators

Description: The exam tips panel advises users to “Flag questions you’re unsure about and come back.” The flag feature works on individual questions (toggle between Flag/Flagged). However, the question number grid at the bottom does not visually distinguish flagged questions from unflagged ones. This breaks the “come back” part of the workflow — users can’t scan the grid to find flagged questions.

Recommended fix: Add a visual indicator to flagged question numbers in the grid. Options include: a small orange/yellow dot overlay on the number button, a different background color, or a border change. The grid already distinguishes three states (unanswered = gray, current = blue, answered = green). Add a fourth: flagged = yellow/orange border or dot.

Implementation detail: The grid buttons are rendered in a map. Add a conditional class:

```
// Pseudo-code for grid button styling
const gridButtonClass = (qIndex) => {
  if (qIndex === currentQuestion) return 'bg-blue-500';
  if (answers[qIndex] !== undefined && flagged[qIndex])
    return 'bg-green-500 ring-2 ring-yellow-400';
  if (answers[qIndex] !== undefined) return 'bg-green-500';
  if (flagged[qIndex]) return 'bg-yellow-500/30 ring-2 ring-yellow-400';
  return 'bg-gray-600';
};
```

3.5 Analytics (#/analytics)

What Was Tested

- Stats cards (Overall Accuracy, Total Questions, Exams Taken, Exam Pass Rate)
- Accuracy by Domain bar chart
- By Difficulty breakdown
- Needs Improvement / Strongest Areas sections
- Data accuracy against actual practice history

Results

Test Case	Expected	Actual	Status
Overall accuracy	50% (1 correct / 2 total)	50% displayed	PASS
Total questions	2	2 displayed	PASS
D1 bar chart	~50% bar for D1, empty for D2-D8	Red bar at ~50 for D1, others empty	PASS
Difficulty breakdown	Medium: 50% (2 questions)	Medium: 50% (2 questions) with yellow bar	PASS
Needs Improvement	D1 at 50%	D1: Security and Risk Management 50%	PASS
Strongest Areas	Should be empty or different from Needs	D1: Security and Risk Management 50%	FAIL

Bug: Same Domain Appears in Both “Needs Improvement” and “Strongest Areas”

Severity: Low (display logic, not data corruption)

Description: When only one domain has been practiced, it appears simultaneously in both the “Needs Improvement” and “Strongest Areas” sections. With a 50% score, Domain 1 is shown in both panels with identical formatting. This is logically contradictory and will confuse users.

Recommended fix:

1. Define a threshold (suggested: 70%) to categorize domains
2. Domains below threshold → Needs Improvement only
3. Domains at or above threshold → Strongest Areas only
4. If fewer than 2 domains have data, show a “Keep practicing to see trends” placeholder instead of forcing single-domain data into both categories
5. Sort Needs Improvement ascending (worst first), Strongest descending (best first)

4. Question Bank Analysis

The current question bank contains 80 questions across 8 domains. This section analyzes distribution, adequacy, and sourcing recommendations.

4.1 Current Distribution

Domain	Name	Exam Weight	Questions	Qs per Weight %	Adequacy
D1	Security and Risk Management	15%	16	1.07	Thin
D2	Asset Security	10%	8	0.80	Thin
D3	Security Architecture and Engineering	13%	12	0.92	Thin
D4	Communication and Network Security	13%	10	0.77	Thin
D5	Identity and Access Management	13%	9	0.69	Thin
D6	Security Assessment and Testing	12%	7	0.58	Critical
D7	Security Operations	13%	10	0.77	Thin
D8	Software Development Security	11%	8	0.73	Thin
TOTAL		100%	80	0.80 avg	

Assessment

80 questions is insufficient for meaningful exam preparation. A typical study session of 25 questions would exhaust nearly a third of the bank. After 3-4 sessions, the user will start seeing repeats frequently, which creates false confidence (memorizing answers rather than understanding concepts).

Target: 250-400 questions total. Distribution should mirror exam weights:

Domain	Current	Target (300 total)	Questions to Add
D1: Security and Risk Mgmt	16	45	29
D2: Asset Security	8	30	22
D3: Security Architecture	12	39	27
D4: Communication/Network	10	39	29

D5: IAM	9	39	30
D6: Security Assessment	7	36	29
D7: Security Operations	10	39	29
D8: Software Development	8	33	25
TOTAL	80	300	220

4.2 Content Sourcing Recommendations

The following are the primary reference materials for building a high-quality CISSP question bank. Questions should be original (not copied) but modeled on the style and cognitive level of these sources.

Primary Study References

- **(ISC)² Official CISSP Study Guide, 10th Edition** (Seidl/Chapple) — The canonical reference. Each chapter maps to a domain. Use concept summaries and review questions as templates for your questions. Focus on the areas where the book goes deeper than your current content: specifically risk quantification formulas (ARO, SLE, ALE, ROSI), legal frameworks by jurisdiction, and cryptographic algorithm selection criteria.
- **(ISC)² Official CISSP Practice Tests, 3rd Edition** (Chapple/Stewart/Gibson) — 1,300+ questions organized by domain. The question style is the closest to the actual exam's cognitive level. Study the stem construction: questions that test "BEST" or "MOST" answers rather than simple recall. Model your new questions after this style.
- **Boson ExSim for CISSP** — Widely considered the hardest CISSP practice questions available. Their explanations for wrong answers are particularly good — they explain WHY each incorrect option fails, not just why the correct one is right. Adopt this pattern for your explanations.

Conceptual / Teaching References

- **Destination Certification MindMap series** (YouTube, free) — Kelly Handerhan's visual walkthroughs of each domain. Excellent for identifying concept connections that cross domains — use these to write questions that test understanding rather than memorization.
- **Luke Ahmed "How to Think Like a Manager for the CISSP Exam"** (free PDF) — The core cognitive framework for CISSP. Every question explanation should include a one-line "CISSP Perspective" note that reinforces managerial thinking. Example: "CISSP Tip: When asked about incident response priorities, the answer is always: protect human life first, then contain the breach."
- **11th Hour CISSP, 4th Edition** (Joshua Feldman, Seth Misenar) — Condensed review. Good for identifying the absolute minimum concepts that must be covered per domain — use as a checklist to verify your question bank has coverage.

Primary Source Documents (for technical accuracy)

- **NIST SP 800-53 Rev 5** — Security and Privacy Controls. Domains 1, 6, 7 heavily reference these control families. Questions should test control selection, not memorization of control IDs.
- **NIST SP 800-37 Rev 2** — Risk Management Framework. Domain 1 questions on RMF steps (Categorize, Select, Implement, Assess, Authorize, Monitor) should reference this directly.
- **NIST SP 800-61 Rev 2** — Computer Security Incident Handling Guide. Domain 7 incident response questions should align with these phases.
- **OWASP Top 10 (2021)** — Domain 8 application security questions should map to current OWASP categories (A01:Broken Access Control through A10:SSRF).
- **ISO/IEC 27001:2022 and 27002:2022** — Information Security Management Systems. Domain 1 governance questions and Domain 6 audit questions reference these standards.

Question Quality Guidelines

Each new question should meet these criteria:

1. **Scenario-based:** Present a situation, then ask what the BEST/MOST/FIRST action is. Avoid pure recall questions (“What does CIA stand for?”).
2. **Four plausible options:** All four answers should be real security concepts. Wrong answers should be wrong for a specific, explainable reason — not obviously absurd.
3. **Managerial perspective:** The correct answer should reflect what a CISO or security manager would choose, not what a technician would implement.
4. **Explanation depth:** Each explanation should cover: why the correct answer is right, why each wrong answer is wrong (specifically), and a CISSP exam tip.
5. **Difficulty calibration:** Easy = single concept recall. Medium = concept application in a scenario. Hard = multi-domain integration or prioritization under constraints.
6. **Reference tagging:** Each question should reference a specific topic (e.g., “Risk Management — Quantitative Analysis”) for analytics granularity.

5. Enhancement Roadmap

Prioritized by impact on learning outcomes and implementation effort.

5.1 P0 — Critical (Do First)

5.1.1 Practice Session Results Screen

Attribute	Detail
Priority	P0 — Critical
Effort	Small (1-2 hours)
Impact	Closes the primary learning feedback loop
Dependencies	None — quiz engine already tracks all needed data

Current behavior: End Session → domain selection screen. No summary.

Target behavior: End Session → results screen showing score, time, missed question review, retry option.

Component specification:

```
interface PracticeResultsProps {
  answers: Array<{
    questionId: number;
    selected: number;
    correct: number;
    isCorrect: boolean;
    domain: number;
    difficulty: string;
  }>;
  totalTime: number; // seconds elapsed
  domain: string;   // filter used
}

// Sections to render:
// 1. Score hero: X/Y correct (Z%) with pass/fail indicator
// 2. Time spent: Mm Ss
// 3. Domain breakdown (if multi-domain session)
// 4. Missed questions list with:
//    - Question text
//    - Your answer (red)
//    - Correct answer (green)
//    - Explanation
// 5. Actions: [Retry Missed] [New Session] [Dashboard]
```

5.1.2 Scale Question Bank to 250+

Attribute	Detail

Priority	P0 — Critical
Effort	Large (10-20 hours of content creation)
Impact	Prevents question memorization; enables meaningful adaptive learning
Dependencies	Content sourcing from references in Section 4.2

See Section 4 for full distribution targets and sourcing guidance. Consider generating questions in batches of 20-30 per domain, reviewing for accuracy, then integrating.

Data format for new questions (must match existing schema):

```
{
  id: number,            // unique, auto-increment
  domain: number,        // 1-8
  difficulty: string,    // 'easy' | 'medium' | 'hard'
  question: string,      // stem text
  options: string[],     // 4 options (A-D)
  correct: number,       // 0-3 (index of correct option)
  explanation: string,   // why correct + why others wrong
  reference: string      // topic tag for analytics
}
```

5.1.3 Fix Study Days / Study Streak Tracking

Attribute	Detail
Priority	P0 — Critical
Effort	Small (30 minutes)
Impact	Motivational feature is visibly broken; undermines trust in tracking
Dependencies	None

Fix: Initialize `studyStartDate` on first question answered. Increment study days by counting distinct calendar dates in `questionHistory` timestamps. Streak = consecutive calendar days with at least one entry, counting backwards from today.

5.2 P1 — High (Do Next)

5.2.1 Spaced Repetition / Adaptive Weighting

Attribute	Detail
Priority	P1 — High
Effort	Medium (4-6 hours)
Impact	Transforms the tool from a static quiz into an adaptive learning system
Dependencies	Requires 200+ questions to be meaningful

Concept: Questions answered incorrectly should reappear sooner and more frequently. The `questionHistory` array already stores the timestamp and correctness of each attempt. Use this to implement a simple SM-2-inspired algorithm:

1. **Track per-question state:** interval (days until next review), easeFactor (multiplier), nextReviewDate
2. **On correct answer:** increase interval (interval * easeFactor), adjust easeFactor up slightly
3. **On wrong answer:** reset interval to 1, decrease easeFactor (min 1.3)
4. **Question selection:** prioritize questions where nextReviewDate <= today, then unseen questions, then random

This doesn't require a backend — all state fits in localStorage. Add a new field to the persisted state:

```
questionSRS: {
  [questionId]: {
    interval: number,      // days
    easeFactor: number,     // starts at 2.5
    nextReview: string,     // ISO date
    repetitions: number     // consecutive correct
  }
}
```

5.2.2 "Think Like a Manager" Tips in Explanations

Attribute	Detail
Priority	P1 — High
Effort	Medium (content creation, 1-2 hours for existing 80 Qs)
Impact	Directly addresses the #1 reason people fail the CISSP
Dependencies	None

The CISSP exam tests managerial judgment, not technical depth. Every explanation should end with a one-line perspective note. Examples:

- *“CISSP Perspective: The exam prioritizes risk management over risk elimination. A control that costs more than the asset it protects is never the right answer.”*
- *“CISSP Perspective: When multiple answers seem correct, choose the one that addresses the business need first, then the technical need.”*
- *“CISSP Perspective: Senior management is always ultimately responsible. Authority can be delegated; responsibility cannot.”*

Schema change: Add a `cisspTip` string field to each question object. Render it below the explanation in a visually distinct callout box (e.g., blue-tinted background with a lightbulb icon).

5.3 P2 — Medium

5.3.1 Flagged Question Indicators in Exam Grid

See detailed specification in Section 3.4. Effort: 30 minutes. Add conditional styling to grid buttons based on flagged state.

5.3.2 Clickable Domain Cards on Dashboard

Each domain card in the progress grid should navigate to `#/practice` with that domain pre-selected. Effort: 30 minutes. Add `onClick={() => navigate('/practice', { domain: domainId })}` and consume the parameter in the Practice component's initial state.

5.3.3 Study Path Task Deep-Linking

Study path tasks that say "Complete 10 practice questions" should link directly to a pre-filtered practice session for that domain. Effort: 1 hour. Add `onClick` handlers to task items that navigate to `#/practice?domain=X`.

5.3.4 Fix Analytics Dual-Display Bug

See Section 3.5 for details. Effort: 30 minutes. Add threshold logic to categorize domains into one section only.

5.4 P3 — Low (Nice to Have)

5.4.1 Mobile Responsive Pass

The sidebar uses `hidden lg:flex` with a hamburger toggle, so the structural foundation exists. However, only 2 `sm:` and 2 `md:` Tailwind breakpoints exist across the entire app. The domain grid, stats cards, question options, and exam grid likely overflow or compress at phone widths. Do a single pass adding responsive classes to these layouts. Effort: 2-3 hours.

5.4.2 Keyboard Navigation in Quiz

Allow A/B/C/D keypresses to select answers, Enter to confirm, and right arrow for next question. Reduces friction for power users doing rapid-fire practice. Effort: 1 hour.

5.4.3 Export/Import Study Data

Since all data is in a single localStorage key, add a Settings page with Export (download `cissp-trainer` as JSON) and Import (upload JSON, validate, merge/replace). Protects against accidental data loss and enables device transfer. Effort: 1-2 hours.

6. Implementation Priority Matrix

Summary of all findings with effort estimates and suggested sprint allocation.

#	Item	Type	Priority	Effort	Sprint
1	Practice results screen	Enhancement	P0	2h	1
2	Fix study days / streak tracking	Bug fix	P0	30m	1
3	Fix analytics dual-display	Bug fix	P2	30m	1
4	Flagged indicators in exam grid	Enhancement	P2	30m	1
5	Clickable domain cards	Enhancement	P2	30m	1
6	Scale question bank to 300	Content	P0	15-20h	2-3
7	Add CISSP tips to explanations	Content	P1	2h	2
8	Spaced repetition engine	Enhancement	P1	4-6h	3
9	Study path task deep-linking	Enhancement	P2	1h	3
10	Mobile responsive pass	Enhancement	P3	2-3h	4
11	Keyboard navigation in quiz	Enhancement	P3	1h	4
12	Export/import study data	Enhancement	P3	1-2h	4

Sprint 1 (quick wins): Items 1-5 = ~4 hours total. Fixes both bugs and delivers the highest-impact enhancement.

Sprint 2-3 (content + intelligence): Items 6-9 = ~22-29 hours. Transforms the tool from a static quiz into an adaptive learning system.

Sprint 4 (polish): Items 10-12 = ~4-6 hours. Quality-of-life improvements.

End of UAT Report